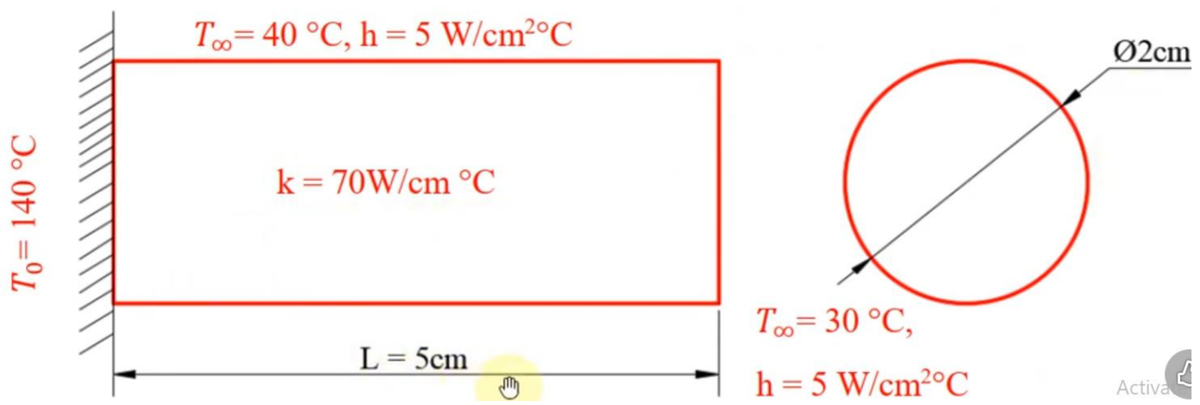
Practical 5

Title – Thermal Analysis – Static Analysis (A)

Sample Example

STATIC THERMAL ANALYSIS OF CIRCULAR FIN

PROBLEM: find the temperature distribution in a straight fin with the physical properties as shown in figure. Thermal conductivity k = 70W/cm °C, convection heat transfer coefficient h = 5W/cm^2°C. temperature at the root of the fin T_0 = 140 °C, Surrounding temperature T_∞= 40 °C.



AIM: To find the temperature distribution in a straight fin.

<u>Procedure</u>

<u>Step 1 Preferences</u>

Open Ansys APDL – Preferences – Thermal – OK

<u>Step 2 Pre-processor</u>

Pre-processor – Element Type – Add – Solid – Tet 10 node 87 - OK - Close

<u>Step 3 Material</u>

Material Properties – Material Models – Thermal – Conductivity – Isotropic – Thermal Conductivity – Enter 70 W/cm °C - OK

Step 4 Modeling

Modeling – Create – Volumes – Cylinder – Solid Cylinder - Enter Radius 1 cm and length 5 cm – OK (select isometric view)

Step 5 Meshing

Meshing – Mesh Tool – Element Attribute – Volumes – Tick smart size – choose fine – (Check mesh – volumes – shape – tetrahedral) - Mesh – select object – OK

Click Raise Hidden – Close Mesh Tool Window

Step 6 Loads

Loads – Define Loads –Apply – Thermal – Temperature – On Areas – Select fixed side of cylinder – Ok – Choose Temperature – Enter Load Temperature Value as 140°C – OK

Loads – Define Loads –Apply – Thermal – Convection – on areas – select top and bottom area of cylinder – OK- Enter Vali Film Coefficient as 5 W/cm °C and enter Vali bulk temperature as 40°C – OK

Loads – Define Loads –Apply – Thermal – Convection – on areas – select free side area of cylinder – OK – enter Vali film coefficient as 5 W/cm °C and enter Vali bulk temperature as 30°C – OK

Step 7 Solution

Solution – Solve –Current LS - OK

Step 8 General Results

General Post Process – Plot Results – Contour Plot – Nodal Solution – DOF – Nodal Temperature - OK

To See Animation

Plot Control – Animate – Deformed Results – Temperature – OK

General Post Process – Plot Results – Contour Plot – Nodal Solution – Thermal Gradient – Thermal Gradient Vector Sum - OK

General Post Process – Plot Results – Contour Plot – Element Solution – Thermal Flux – Thermal Flux Vector Sum - OK

General Post Process – Plot Results – Contour Plot – Element Solution – Heat Flow – OK

General Post Process – List Results – Nodal Solution – Nodal Temperature - OK